Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/12/2007
PI Name:	Chen, Lea Der Ph.D.		
Project Title:	Reflight of ELF (Enclosed Laminar Flames) Investigation		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous flames - premixed		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ldchen@engineering.uiowa.edu	**Fax:**	FY 319-384-0576
PI Organization Type:	UNIVERSITY	**Phone:**	319-335-5674
Organization Name:	The University of Iowa		
PI Address 1:	3131 SC, Deparment of Mechanical & Industrial Engineering		
PI Address 2:	The University of Iowa		
PI Web Page:			
City:	Iowa City	**State:**	IA
Zip Code:	52242	**Congressional District:**	2
Comments:	NOTE (June 2016): Dr. Chen is currently at Texas A&M Corpus Christi		
Project Type:	Ground	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	01/01/1999	**End Date:**	01/03/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	4
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 1/03/2007, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Stocker, D.P. (NASA Glenn Research Center) Brooker, J.E. (NASA Glenn Research Center)		
Grant/Contract No.:	NCC3 666		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:	The overall objective of the program is to study the burner stability of enclosed laminar flames in microgravity environments. The specific objectives are (a) to evaluate the effects of fuel Schmidt number on the transition from lifted flames to blow-out conditions, (b) to study the hysteresis effects observed in the reattachment of lifted flames, (c) to gain physical insight into the transition from lifted to blow-out conditions, and (d) to validate numerical simulation. Ground-based experiments are conducted to support the space experiments The current project studies the flame stabilization in coflow environments. The findings will contribute to the fundamentals of combustion theory. The current invetsigation is also of practical significance as practical combustors often have jet flames in flow environments, for example, co-flow inside gas turbine engine combustors and in industrial furnaces. Understanding of the flame stabilization at the fundamental level is key to safety considerations in the design of new combustors. The verified numerical model for the prediction of the flame stabilization can be applied as a design tool for complex engineering systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This invetsigation provides additional insight of the structure of the edge flame from stabilized to (near) blowout conditions. Specifically, the ratio of length scales versus the ratio of time scales was extracted from the numerical results, and the ratio of length scales versus Damkohler number was used to characherize the flame regimes. The adpative mesh refinement flame code will also be used to study the deNOx process in catalytic converters.
Task Progress:	An investigation of buoyancy effects on the structure and stability of Burke-Schumann diffusion flames (BSDF) is conducted. The specific objectives are to (1) investigate the importance of the buoyancy-dependent flow-field as affected by oxidizer flow on flame stabilization and (2) examine the state relationships (i.e., flame structure.) of co-flow diffusion flames under the influence of buoyancy conditions. The goal of the research is to improve our understanding of the influence of buoyancy on the stability and structure of Burke-Schumann type diffusion flames. The publications including student theses that result from this project are summarized in the report. A new numerical method, Lattice Boltzmann Method, was successfully used for simulation of non-premixed flames as documented in the article by Lee et al. (2006) cited in Bibliography.
Bibliography Type:	Description: (Last Updated: 06/13/2007)
Articles in Peer-reviewed Journals	Lee TH, Lin C-L, Chen L-D. "A lattice Boltzmann algorithm for calculation of the laminar jet diffusion flame," Journal of Computational Physics 2006 Jun 10; 215(1): 133-52. http://dx.doi.org/10.1016/j.jcp.2005.10.021 , Jun-2006
Articles in Peer-reviewed Journals	Venuturumilli R, Chen L-D. "Numerical simulation of laminar jet diffusion." Numerical Heat Transfer B: Fundamentals 2004 Aug;46(2):101-20. http://dx.doi.org/10.1080/10407790490449507 , Aug-2004
Dissertations and Theses	Venuturumilli R. "Numerical Simulation of Non-Premixed Laminar Air Jet Diffusion Flames Using Adaptive Mesh Refinement." Ph.D. Dissertation, The University of Iowa, Iowa City, Iowa, 2005. , Dec-2005
Dissertations and Theses	Zhang Y. "Experimental Study of Stability of Laminar Nonpremixed Flame in Coflowing Air." M.S. Thesis, The University of Iowa, Iowa City, Iowa, 2003. , Dec-2003